

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3

Total Marks:30

Annexure C

Case study

Criteria	Understanding of Case (2.5)	Application of Concepts (2.5)	Analysis (2)	Solution Design (2)	Clarity (1)	Total(10)
Weightage	25%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						

Code of Conduct:

*I 192472115(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:U.Midhula

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANNEXURE A

Case Study:

1. A shepherd boy gets bored tending the town's flock. To have some fun he cries out, "Wolf!" even though no wolf is in sight. The villagers run to protect the flock, but then get really mad when they realize the boy was playing a joke on them.

One night, the shepherd boy sees a real wolf approaching the flock and calls out, "Wolf!" The villagers refuse to be fooled again and stay in their houses. The hungry wolf turns the flock into lamb chops. The town goes hungry. Panic ensues. Identify TP,FP,FN,TN.

Ans:

Yes. This is **Assessment Tool 2**, and it has 3 questions for 30 marks. I can prepare it in the same style as Assessment Tool 1: **step-by-step calculations, Weka procedure where required, analysis, and a final answer that you can directly paste into Google Docs.**

I'll start with **Q1**.

Q1. Shepherd Boy Case Study – Confusion Matrix

Given Situation

The shepherd boy calls “**Wolf!**” when there is no wolf as a joke. Later, when a real wolf approaches, he calls “Wolf!” again, but the villagers do not respond.

We can treat:

- **Actual condition:** Whether a wolf is really present
- **Prediction:** Whether the boy reports “Wolf!”

Classification

Actual Situation	Boy's Call	Classification
Wolf present	“Wolf!”	TP
No wolf	“Wolf!”	FP

Wolf present No “Wolf!” **FN**

No wolf No “Wolf!” **TN**

Identifying TP, FP, FN and TN

1. True Positive (TP)

A wolf is actually present and the boy correctly calls:

“Wolf!”

Therefore:

TP={Real wolf present and boy cries wolf}

2. False Positive (FP)

There is no wolf, but the boy cries:

“Wolf!”

This happens when he is playing a joke on the villagers.

FP {No wolf present but boy cries wolf}

3. False Negative (FN)

A wolf is actually present, but the boy does not successfully report it / the warning is not acted upon.

For classification purposes:

FN={Wolf present but classified/reported as no wolf}

4. True Negative (TN)

There is no wolf and the boy does not cry wolf.

TN={No wolf present and no wolf is reported}

Confusion Matrix

	Actual Wolf	Actual No Wolf
Predicted Wolf	TP	FP
Predicted No Wolf	FN	TN

Final Answer

Term **Meaning in the case**

TP Wolf is present and boy cries “Wolf!”

FP No wolf is present but boy cries “Wolf!”

FN Wolf is present but boy does not correctly report “Wolf!”

TN No wolf is present and boy does not cry “Wolf!”

Analysis

The case demonstrates the importance of distinguishing between correct and incorrect predictions. The boy's false alarms represent **False Positives**, because villagers are warned about a wolf when there is actually no wolf. Repeated false positives reduce the villagers' trust in the warning system. Consequently, when the real wolf appears, the warning is ignored, creating the practical effect of a **False Negative**.

Conclusion

The shepherd boy example illustrates the four basic outcomes of a classification system: TP, FP, FN, and TN. False positives are particularly important in this case because repeated false alarms cause people to ignore a genuine warning later. This demonstrates why classification systems must balance correct detection with minimizing false alarms.

2. Assume there are 100 images, 30 of them depict a cat, the rest do not. A machine learning model predicts the occurrence of a cat in 25 of 30 cat images. It also predicts absence of a cat in 50 of the 70 no cat images. Identify TP, FP, FN, TN and Calculate the Precision and Recall and F1 Score .

Ans:

Given Data

There are **100 images** in total.

- Cat images = **30**
- No-cat images = **70**

The machine learning model:

- Correctly predicts a cat in **25 of the 30 cat images**
- Correctly predicts no cat in **50 of the 70 no-cat images**

We need to find:

TP, FP, FN, TN, Precision, Recall and F1 Score.

1. Identify TP, FN, TN and FP

True Positive (TP)

There are 30 actual cat images.

The model correctly identifies 25 as cat:

TP=25

False Negative (FN)

Out of 30 actual cat images, 25 were correctly identified.

Therefore:

FN=30-25

FN=5

These are cat images that the model incorrectly classified as no cat.

True Negative (TN)

There are 70 images without cats.

The model correctly identifies 50 as no cat:

$$TN=50$$

False Positive (FP)

Out of 70 no-cat images, 50 were correctly identified.

Therefore:

$$FP=70-50$$

$$FP=20$$

These are images without cats that the model incorrectly classified as containing a cat.

2. Confusion Matrix

	Actual Cat	Actual No Cat
Predicted Cat	TP = 25	FP = 20
Predicted No Cat	FN = 5	TN = 50

Check:

$$25+20+5+50=100$$

Therefore, all 100 images are accounted for.

3. Calculate Precision

Precision measures how many of the images predicted as **cat** were actually cats.

Formula:

$$\text{Precision} = \frac{\{TP\}}{\{TP+FP\}}$$

Substitute the values:

$$\text{Precision} = \frac{\{25\}}{\{25+20\}}$$

$$= \frac{\{25\}}{\{45\}}$$

$$\text{Precision} = 0.5556$$

Percentage:

$$\text{Precision} = 55.56\%$$

Interpretation

When the model predicts that an image contains a cat, it is correct approximately **55.56%** of the time.

4. Calculate Recall

Recall measures how many of the actual cat images were correctly identified.

Formula:

$$\text{Recall} = \frac{\{TP\}}{\{TP+FN\}}$$

Substitute:

$$\text{Recall} = \frac{\{25\}}{\{25+5\}}$$

$$= \{25\} / \{30\}$$

$$\text{Recall} = 0.8333$$

Percentage:

$$\text{Recall} = 83.33\%$$

Interpretation

The model successfully detects approximately **83.33%** of all actual cat images.

5. Calculate F1 Score

Formula:

$$F1 = 2 \{ \text{Precision} \times \text{Recall} \} / \{ \text{Precision} + \text{Recall} \}$$

Substitute:

$$F1 = 2 \{ 0.5556 \times 0.8333 \} / \{ 0.5556 + 0.8333 \}$$

$$F1 = \text{approx} 0.6667$$

Therefore:

$$F1 = 0.6667$$

Percentage:

$$F1 = 66.67\%$$

6. Final Results

Measure	Calculation	Result
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TP	Correctly predicted cats	25
FP	No-cat predicted as cat	20
FN	Cat predicted as no-cat	5
TN	Correctly predicted no-cat	50
Precision	$25/(25+20)$	55.56%
Recall	$25/(25+5)$	83.33%
F1 Score	$2(PR)/(P+R)$	66.67%

Analysis of Results

The model correctly identifies **25 out of 30 cat images**, giving a **recall of 83.33%**. This indicates that the model is effective at detecting actual cat images and misses only 5 cat images.

However, the model produces **20 false positives**, where images without cats are incorrectly classified as containing cats. As a result, the **precision is only 55.56%**. This means that nearly half of the images predicted as cats are actually non-cat images.

The **F1 score of 66.67%** represents the balance between precision and recall. The model has relatively high recall but lower precision, indicating that it prioritizes detecting cats but generates a considerable number of false alarms.

Conclusion

The classification model has a **recall of 83.33%, precision of 55.56%, and F1 score of 66.67%**. The model is good at detecting actual cat images but needs improvement in reducing false positive predictions. Therefore, improving precision while maintaining the current recall would make the model more effective.

3. An email service provider wants to classify emails as **Spam or Not Spam** using features:

- Contains “Offer” (Yes/No)

- Contains “Free” (Yes/No)
- Email Length (Short/Long)

Sample Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam
No	No	Short	Not Spam

Questions

1. Calculate prior probabilities:
 - o $P(\text{Spam})$, $P(\text{Not Spam})$
2. Compute conditional probabilities for each feature.
3. Using **Naïve Bayes**, classify:
 - o (Offer=Yes, Free=No, Length=Short)
4. Compare results using **WEKA (Naïve Bayes classifier)**.
5. Discuss the **independence assumption** in Naïve Bayes

Ans:

Yes. For **Q3, Weka is not necessary**. We can answer the entire question using **manual Naïve Bayes calculations**. You can directly paste the following into your assessment.

Q3. Email Classification Using Naïve Bayes

Given Dataset

Offer	Free	Length	Class
Yes	Yes	Short	Spam
Yes	No	Long	Not Spam
No	Yes	Short	Spam
No	No	Long	Not Spam
Yes	Yes	Long	Spam

No No Short Not Spam

Total number of emails:

[
N=6
]

There are:

- Spam = 3
- Not Spam = 3

1. Prior Probabilities

Probability of Spam

$P(\text{Spam}) = \{\text{Number of Spam emails}\} / \{\text{Total emails}\}$

$P(\text{Spam}) = \{3\} / \{6\}$

$P(\text{Spam}) = 0.5$

Probability of Not Spam

$P(\text{NotSpam}) = \{3\} / \{6\}$

$P(\text{NotSpam}) = 0.5$

Therefore:

Class	Frequency	Prior Probability
Spam	3	0.5

Not Spam 3 **0.5**

2. Conditional Probabilities

For Spam

There are three Spam emails:

Offer	Free	Length
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Yes	Yes	Short
-----	-----	-------

No	Yes	Short
----	-----	-------

Yes	Yes	Long
-----	-----	------

Offer

$$P(\text{Offer}=\text{Yes}|\text{Spam})=\{2\}/\{3\}$$

$$P(\text{Offer}=\text{No}|\text{Spam})=\{1\}/\{3\}$$

Free

$$P(\text{Free}=\text{Yes}|\text{Spam})=\{3\}/\{3\}=1$$

$$P(\text{Free}=\text{No}|\text{Spam})=\{0\}/\{3\}=0$$

Length

$$P(\text{Short}|\text{Spam})=\{2\}/\{3\}$$

$$P(\text{Long}|\text{Spam})=\frac{1}{3}$$

For Not Spam

There are three Not Spam emails:

Offer	Free	Length
Yes	No	Long
No	No	Long
No	No	Short

Offer

$$P(\text{Offer}=\text{Yes}|\text{NotSpam})=\frac{1}{3}$$

$$P(\text{Offer}=\text{No}|\text{NotSpam})=\frac{2}{3}$$

Free

$$P(\text{Free}=\text{Yes}|\text{NotSpam})=\frac{0}{3}=0$$

$$P(\text{Free}=\text{No}|\text{NotSpam})=\frac{3}{3}=1$$

Length

$$P(\text{Short}|\text{NotSpam})=\frac{1}{3}$$

$$P(\text{Long}|\text{NotSpam})=\{2\}/\{3\}$$

Conditional Probability Table

Feature	Value	Spam	Not Spam
Offer	Yes	2/3	1/3
Offer	No	1/3	2/3
Free	Yes	1	0
Free	No	0	1
Length	Short	2/3	1/3
Length	Long	1/3	2/3

3. Classify the New Email

The new email has:

- **Offer = Yes**
- **Free = No**
- **Length = Short**

Let:

$$X=(\text{Offer}=\text{Yes},\text{Free}=\text{No},\text{Length}=\text{Short})$$

Using the Naïve Bayes formula:

$$P(\text{Class}|X)\propto P(\text{Class})\times P(\text{Offer}|\text{Class})\times P(\text{Free}|\text{Class})\times P(\text{Length}|\text{Class})$$

Calculate for Spam

$$\begin{aligned} &P(\text{Spam}|X) \propto \\ &P(\text{Spam}) \\ &\times P(\text{Offer}=\text{Yes}|\text{Spam}) \\ &\times P(\text{Free}=\text{No}|\text{Spam}) \\ &\times P(\text{Short}|\text{Spam}) \end{aligned}$$

Substitute:

$$=0.5 \times \frac{2}{3} \times 0 \times \frac{2}{3}$$

$$\{P(\text{Spam}|X)=0\}$$

Calculate for Not Spam

$$\begin{aligned} &P(\text{NotSpam}|X) \propto \\ &P(\text{NotSpam}) \\ &\times P(\text{Offer}=\text{Yes}|\text{NotSpam}) \\ &\times P(\text{Free}=\text{No}|\text{NotSpam}) \\ &\times P(\text{Short}|\text{NotSpam}) \end{aligned}$$

$$=0.5 \times \frac{1}{3} \times 1 \times \frac{1}{3}$$

$$=\frac{1}{18}$$

$$\{P(\text{NotSpam}|X)=0.0556\}$$

Since:

$$0.0556 > 0$$

the email is classified as:

{NOT SPAM}

4. Zero-Frequency Problem

The calculation gives:

$$P(\text{Free}=\text{No}|\text{Spam})=0$$

because none of the Spam emails in the given training data has **Free = No**.

This is known as the **zero-frequency problem** in Naïve Bayes.

To overcome this problem, **Laplace smoothing** can be used.

The Laplace smoothing formula is:

$$P(\text{value}|\text{class}) = \frac{\text{count}(\text{value}, \text{class}) + 1}{\text{count}(\text{class}) + k}$$

where (k) is the number of possible values.

For example:

$$P(\text{Free}=\text{No}|\text{Spam})$$

$$\frac{\{0+1\}}{\{3+2\}}$$

$$=\frac{\{1\}}{\{5\}}=0.2$$

Even after applying smoothing, the final classification is **Not Spam**.

5. Independence Assumption

Naïve Bayes assumes that the features are **conditionally independent given the class**.

Therefore:

$$P(\text{Offer}, \text{Free}, \text{Length} | \text{Class})$$

is calculated as:

$$\begin{aligned} &P(\text{Offer} | \text{Class}) \\ &\times P(\text{Free} | \text{Class}) \\ &\times P(\text{Length} | \text{Class}) \end{aligned}$$

For example:

$$P(\text{Offer}=\text{Yes}, \text{Free}=\text{No}, \text{Length}=\text{Short} | \text{Spam})$$

is calculated as:

$$P(\text{Offer}=\text{Yes} | \text{Spam})$$

$$P(\text{Free}=\text{No} | \text{Spam})$$

$$P(\text{Short} | \text{Spam})$$

In real-world emails, these features may not be completely independent. For example, an email containing **"Offer"** may also be more likely to contain **"Free"**. Therefore, the independence assumption may not always hold perfectly.

However, Naïve Bayes is still widely useful because it is **simple, fast, and effective for classification**.

Analysis of Results

The Naïve Bayes classifier was used to classify an email as **Spam or Not Spam** based on the features Offer, Free, and Email Length. The dataset contains six emails, with three Spam and three Not Spam emails. Hence, both classes have an equal prior probability of **0.5**.

For the test email with **Offer = Yes, Free = No, and Length = Short**, the calculated probability for Spam becomes zero because no Spam email in the training dataset contains **Free = No**. The probability for Not Spam is **0.0556**, which is greater than the Spam probability. Therefore, the test email is classified as **Not Spam**.

The zero probability demonstrates the **zero-frequency problem** of Naïve Bayes. Laplace smoothing can be applied to avoid this problem by assigning a small non-zero probability to previously unseen feature combinations.

Conclusion

The Naïve Bayes classification result for the given email is **Not Spam**. The calculation demonstrates how prior probabilities and conditional probabilities are combined to make a prediction. The analysis also shows the importance of Laplace smoothing when a feature value has not appeared in a particular class. Although the independence assumption may not always be completely true in real-world email data, Naïve Bayes remains a simple and efficient classification technique.